AP BIOLOGY TOPIC 1.4

Carbohydrates: Monosaccharides & Polysaccharides

UNIT 1: CHEMISTRY OF LIFE (8–11% EXAM WEIGHT)

SYI-1.B | GLYCOSIDIC LINKAGES | A VS. B GLUCOSE

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 1.4 (CARBOHYDRATES: MONOSACCHARIDES & POLYSACCHARIDES)

CED Core Vocabulary

Smallest Carbohydrate Units — Monomers with general empirical formula $(CH_2O)_n$ (carbon:hydrogen:oxygen in a 1 : 2 : 1 ratio).

Aldose — (carbonyl at end of carbon chain) → e.g., Glucose, Galactose .

Isomers — Molecules with the same molecular formula ($C_6H_{12}O_6$) but different covalent or spatial arrangements → leads to distinct biological recognition by enzymes....

Maltose — (malt sugar) = Glucose + Glucose

Chemical Structure — Carbon Backbone: 3 to 7 carbons (trioses = 3C like G3P; pentoses = 5C like ribose/deoxyribose; hexoses = 6C like glucose).

Ketose — (carbonyl in middle of carbon chain) → e.g., Fructose .

Stereoisomers (Geometric / Optical) — Same covalent partnerships, different 3D spatial orientation of –OH groups (e.g., glucose vs. galactose at C4; α-glucose vs. β-glucose at C1).

Sucrose — (table sugar / phloem transport) = Glucose + Fructose

§2. MONOSACCHARIDES (SIMPLE SUGARS)

$(CH_2O)_n$ • 1:2:1 Ratio

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

Monosaccharides (Simple Sugars) ➔ **Smallest Carbohydrate Units** ➔ **Chemical Structure** ➔ **ring structures**

- **Smallest Carbohydrate Units:** Monomers with general empirical formula **$(CH_2O)_n$** (carbon:hydrogen:oxygen in a **1 : 2 : 1 ratio**).
- **Chemical Structure:**
  - **Carbon Backbone:** 3 to 7 carbons (trioses = 3C like G3P; pentoses = 5C like ribose/deoxyribose; hexoses = 6C like glucose).
  - **Multiple Hydroxyl Groups (–OH):** Make sugars polar, hydrophilic, and **highly water-soluble** via H-bonding.
  - **One Carbonyl Group (C=O):**
    - **Aldose** (carbonyl at end of carbon chain) → e.g., **Glucose, Galactose**.
    - **Ketose** (carbonyl in middle of carbon chain) → e.g., **Fructose**.
- In aqueous solutions, 5C and 6C sugars spontaneously cyclize into stable **ring structures**.

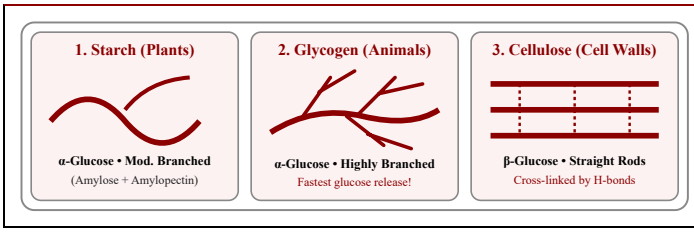
[EXAMPLE & BIOLOGICAL APPLICATION] **Glucose, fructose, and galactose** are all hexose monosaccharides with the identical molecular formula **$C_6H_{12}O_6$**.

§4. POLYSACCHARIDES: STRUCTURE DETERMINES FUNCTION

Complex Carbs

- **Polysaccharides:** Long polymers of hundreds to thousands of monosaccharides joined by covalent **glycosidic linkages** (**1→4** linear chains or **1→6** branch points) formed via dehydration synthesis.
- **Branching vs. Linearity Dictates Biological Role:**
  - **Linear Chains (β-1,4 linkages):** Form straight, unbranched rigid rods where parallel chains cross-link via **thousands of inter-chain hydrogen bonds** → high tensile strength for **structural support**.
  - **Helical & Branched Chains (α-1,4 + α-1,6 linkages):** Compact, bushy helical granules → **osmotically inert energy storage** with many free non-reducing ends for rapid enzymatic hydrolysis.

[AP EXAM FRQ SCORING STRATEGY] **Keyword Alert:** Always state that **glycosidic bond = covalent sugar–sugar linkage**, and that **more branches = more free ends for enzymes to hydrolyze glucose simultaneously → faster ATP mobilization**!



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§3. ISOMERS & DISACCHARIDES (SAME FORMULA, DIFFERENT STRUCTURE)

Specificity

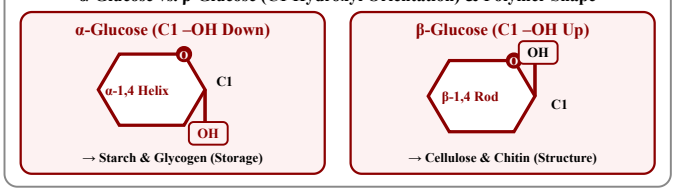
HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

Isomers & Disaccharides (Same Formula, Different Structure) ➔ **Isomers** ➔ **Common Disaccharides ($C_{12}H_{22}O_{11}$)**

- **Isomers:** Molecules with the **same molecular formula** (**$C_6H_{12}O_6$**) but **different covalent or spatial arrangements** → leads to distinct biological recognition by enzymes.
  - **Structural Isomers:** Different covalent bonding order (e.g., **glucose** [aldose] vs. **fructose** [ketose]).
  - **Stereoisomers (Geometric / Optical):** Same covalent partnerships, different 3D spatial orientation of **–OH** groups (e.g., **glucose vs. galactose** at C4; **α-glucose vs. β-glucose** at C1).
- **Common Disaccharides (**$C_{12}H_{22}O_{11}$**):** Formed by dehydration synthesis of 2 monosaccharides:
  - **Maltose** (malt sugar) = Glucose + Glucose
  - **Sucrose** (table sugar / phloem transport) = Glucose + Fructose
  - **Lactose** (milk sugar) = Glucose + Galactose

[EXAMPLE & BIOLOGICAL APPLICATION] **Glucose vs. Galactose** differ only in the 3D orientation of a single **–OH** group on carbon-4, yet require completely different enzymes for metabolism!

α-Glucose vs. β-Glucose (C1 Hydroxyl Orientation) & Polymer Shape



§5. STORAGE VS. STRUCTURAL POLYSACCHARIDES

4 Major Polymers

POLYSACCHARIDE	MONOMER & BOND	BRANCHING & BIOLOGICAL ROLE
Starch *(Plants)*	**α-glucose (** **α-1,4** **)**	**Unbranched (amylose)** or **moderately branched (amylopectin)**; energy storage in plastids/roots.
Glycogen *(Animals)*	**α-glucose (** **α-1,4 & α-1,6** **)**	**Extensively branched**; stored in liver & skeletal muscle for rapid glucose mobilization.
Cellulose *(Plants)*	**β-glucose (** **β-1,4** **)**	**Unbranched straight chains** (every 2nd β-glucose flipped 180°); H-bonded microfibrils in **plant cell walls**.
Chitin *(Fungi/Arthropods)*	**β-N-acetylglucosamine**	Contains **Nitrogen (N)**; structural support in **fungal cell walls** & **arthropod exoskeletons**.

[EXAMPLE & BIOLOGICAL APPLICATION] **Digestibility & Enzyme Specificity:** Humans possess salivary/pancreatic *α-amylase* to hydrolyze **α-1,4** bonds in **starch & glycogen**, but cannot hydrolyze **β-1,4** linkages in **cellulose** (dietary fiber). Herbivores (cows, termites) rely on symbiotic gut microbes producing *cellulase*.

[COMMON AP EXAM FRQ TRAP] **Mixing α vs. β glucose:**
- **α-glucose** → Helical & branched **energy storage** (Starch & Glycogen)
- **β-glucose** → Straight, H-bonded **structural fibers** (Cellulose & Chitin)

[MECHANISTIC NOTE] **AP CED Scope Note:** Memorizing the exact carbon-by-carbon molecular structures of specific carbohydrate polymers is beyond the scope of the AP Exam — focus on **monomer orientation, branching vs. linearity, and functional consequences**!

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 1.4 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 1.4: Carbohydrates: Monosaccharides & Polysaccharides ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Lab Context: Macromolecule Indicator Assays (Benedict's for reducing sugars, Lugol's Iodine for starch, Biuret for peptide bonds) & Buffer Titration Curves. Independent Variable (IV): pH or temperature treatment; Dependent Variable (DV): absorbance/colorimetric change or structural stability; Negative Control: deionized H_2O. **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative & Graph Analysis: Stoichiometric monomer-to-polymer condensation: for n monomers joined in a linear chain, (n − 1) covalent bonds form and (n − 1) H_2O molecules are released. pH scale is logarithmic: $pH = -\log_{10}[H^+]$; each 1-unit decrease in pH represents a 10-fold increase in $[H^+]$. Caloric density: Lipids (~9 kcal/g) vs. Carbohydrates/Proteins (~4 kcal/g). **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Molecular Disruption Case Study: Missense mutation in the β-globin gene (Glu6Val in Sickle-Cell Disease) replaces a polar/charged glutamic acid R-group with a nonpolar hydrophobic valine on the protein exterior, driving abnormal quaternary hydrophobic aggregation into rigid fibers under deoxygenated conditions. Similarly, extreme fever (>40°C) or acidosis (pH < 7.0) disrupts weak hydrogen and ionic bonds in tertiary (3°) protein structure while leaving primary (1°) peptide bonds intact.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 1.4 to prove why the claim holds.

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